



## **School of Computer Science and Engineering**

### **BCSE302P - Database Systems**

#### **Exercise 1**

**Aim: To study Data Definition and Data Manipulation commands.**

Consider a schema with minimum of 3 relations. These 3 relations should have connected through a reference key. Each relation should have attributes of different data types like number, varchar, date etc.

Use SQL queries to perform following operations.

1. Create tables
2. Change data type of a column in a relation
3. Rename a column in a relation
4. Drop a column in a relation
5. Increase size of a column in a relation
6. Rename a relation
7. Insert values in each table (minimum of 5 records in each table)
8. Update values in a relation
9. Delete rows in a relation
10. Retrieve all rows from a relation
11. Retrieve certain column from a relation
12. Retrieve certain rows from a relation
13. Create copy of a table
14. Truncate the contents of the copy table
15. Drop the copy table.

## Exercise: II

Aim: To use constraints and formulate the tables to have consistent data.

Apply below constraints over the relations and ensure that consistency is maintained in the data.

1. Set primary key in every relation.
2. Set a referential integrity constraint (foreign key).
3. Set a not null constraint on a column.
4. Use check constraint and ensure only certain allowed values are entered in a specific column.
5. Set a column as unique.

### **Exercise: III**

Aim: To understand types of function in SQL

Execute the following queries based on the schema specified in exercise 1

1. Apply SQL queries to retrieve data by using all numeric functions.
2. Apply SQL queries to retrieve data by using all character functions.
3. Apply SQL queries to retrieve data by using all date functions.
4. Apply SQL queries to retrieve data by using all conversion functions.

## Exercise: IV

Aim: To understand different operators and group functions in SQL

Execute the following queries based on the schema specified in exercise 1

1. Apply SQL queries to retrieve data by using all arithmetic, logical, set and comparative operators
2. Apply SQL queries to retrieve data by using all aggregate functions.
3. Apply SQL queries to retrieve data by using all aggregate functions and applying a condition over the aggregate function.

## Exercise: V

### Sub Query and View

Aim: to understand the concept of Sub queries and logical tables in oracle

Execute the following queries based on the schema specified in exercise 1

1. Use single row subqueries by applying all the comparison operators
2. Use multiple row subqueries by applying all the operators (in, not in, all and any)
3. Apply a correlated subquery on the schema and retrieve relevant values.
4. Create a simple view.
5. Create a view using subquery.

## Exercise: VI

### Joins

Aim: To understand about retrieving data across multiple tables.

Execute the following queries based on the schema specified in exercise 1

1. Retrieve values from 2 tables using a single join condition.
2. Retrieve values using natural join.
3. Fetch data from two or more tables using left outer join.
4. Fetch data from tables using right outer join.
5. Fetch data from tables using full outer join.

Create the below tables and insert values.

Doctor ( Doctor Id, Name, department, experience)

Patient ( Patient Id, Name, Doc Id, Admitted date, discharge date, disease)

Use SQL to answer the below queries.

1. Display the doctor details along with their patient details. Doctors without any patient should also be listed in the output.
2. Display the patient name and patient\_id who were admitted more than a month in the hospital and serviced by a doctor in ophthalmology department.
3. Retrieve the details of doctor who have more than 2 patients.